

Results

Generated catalog, metrics, and outputs I

The live registry reports 17 implemented generators, 18 catalog entries, 18 evidence records, and 36 source records in the checked-in scholarship audit snapshot. The complete catalog is generated in Appendix A rather than copied into prose. The companion figure keeps the categorical registry, source namespace, and implementation statuses visible at a glance.

Generated catalog, metrics, and outputs II

Catalog status and evidence matrix

Taxonomy facets are orthogonal; status and evidence are separate dimensions.

ID	modality	mechanism	signature	input	status
visual.duck_rabbit	visual	ambiguity	bistability	none	implemented
visual.simultaneous_contrast	visual	contrast_and_context	contrast_distortion	none	implemented
visual.apparent_motion	visual	temporal_motion	illusory_motion	none	implemented
audio.shepard_tone	auditory	spectral_harmonic	continuity	none	implemented
audio.missing_fundamental	auditory	spectral_harmonic	filling_in	none	implemented
audiovisual.sound_induced_flash	audio_visual	crossmodal_temporal	fusion_or_fission	controlled_playback	implemented
audiovisual.ventriloquist	audio_visual	crossmodal_spatial	spatial_capture	stereo_playback	implemented
visual.muller_lyer	visual	geometric_alignment	geometric_distortion	none	implemented
visual.poggendorff	visual	geometric_alignment	geometric_distortion	none	implemented
visual.ponzo	visual	geometric_alignment	geometric_distortion	none	implemented
visual.kanizsa_triangle	visual	contrast_and_context,geometric_all	filling_in	none	implemented
visual.ebbinghaus	visual	contrast_and_context,geometric_all	geometric_distortion	none	implemented
visual.zollner	visual	geometric_alignment	geometric_distortion	none	implemented
audio.tritone_paradox	auditory	spectral_harmonic	categorical_recoding	none	implemented
audio.octave_illusion	auditory	stream_segregation	fusion_or_fission	stereo_playback	implemented
audio.auditory_continuity	auditory	stream_segregation	continuity	none	implemented
audiovisual.temporal_ventriloquism	audio_visual	crossmodal_temporal	temporal_binding_or_recalibration	controlled_playback	implemented
audiovisual.mcquirk	audio_visual	speech_categorization	categorical_recoding	validated_speech_fixture	input_required

Typed domains and delivery profiles I

Table 1: Typed parameter domains and units.

Type	Unit	Validated domain	Role
PixelDimension	pixels	8–4096	image/video width and height
GrayscaleLevels	levels	2–256	representable grayscale levels
QuantizationLevels	levels	2–256	output quantization buckets
FrequencyHz	Hz	> 0 and finite	oscillator and harmonic frequencies
SampleRate	samples/s	1,000–384,000	audio sampling clock
FrameRate	frames/s	1–240	video presentation clock
SyncOffsetMs	ms	–10,000–10,000	audio relative to video
SpatialOffset	normalized	–1–1	declared crossmodal spatial discrepancy

Typed domains and delivery profiles II

The parameter contracts are summarized in [tbl:parameters]. They constrain physical construction and serialization; they are not psychophysical scales.

Table 2: Encoding profiles and backend capabilities.

Format	Backend	Artifact	Profile	Status
PNG	Pillow	image	lossless uint8 raster	available
WAV	python wave	audio	PCM 8/16/24/32-bit	available
GIF	Pillow	video	palette animation with declared duration	available
NPZ	NumPy	canonical artifact	exact little-endian float32 archive	available
MP4	ffmpeg	video/audiovisual	H.264 or muxed MP4; decode inspected	available

Typed domains and delivery profiles III

The delivery profiles are summarized in [tbl:encodings]. Backend availability is environment-dependent and is recorded at generation time.

The catalog distinguishes a physical construction from an observer claim. For example, the sound-induced-flash generator establishes one flash and one or two beep events on a shared clock; it does not establish a reported flash count. Similarly, the temporal-ventriloquism generator exposes a declared timing offset, while the literature reports observer-dependent temporal judgments under specific tasks and timing conditions [vroomen2004temporal].

Pipeline and provenance I

DuckRabbit reproducibility contract

Canonical identity is established before delivery; decoded inspection tests what was actually written.



Engineering trace only: deterministic media facts and verification are not observer evidence.

Figure 2: Five labeled stages connect a typed request to a deterministic canonical artifact, encoded media, decoded inspection, and a verification manifest.

Pipeline and provenance II

What this figure shows: Five labeled stages connect a typed request to a deterministic canonical artifact, encoded media, decoded inspection, and a verification manifest. This schematic separates a typed request from the canonical artifact it generates, the optional delivery encoder, decoded inspection, and the manifest-level verification decision. Arrows indicate data and provenance dependencies rather than a causal model of perception; the seed is 0 and the complete stage record is in source data output/data/architecture.json. Controls: typed request $q=(i, s, e)$ with seed $s=0$; default generator and delivery-independent canonicalization. Objective facts: no unit-bearing objective quantity is applicable to this architecture diagram; stage identities and provenance fields are recorded in the sidecar. Claim level: canonical_stimulus. Source data: output/data/architecture.json (SHA-256 digest recorded in the figure registry). Evidence lineage: formalism:eq:typed_request, formalism:eq:canonical_generation, formalism:eq:encoding_verification. Limitations: The schematic abstracts implementation boundaries and does not represent an observer study or a causal theory of perception. Boundary: The

Visual constructions and parameter sweeps I

Visual constructions and parameter space II

Complete panel of implemented visual constructions

One typed default per implemented entry. Static families show canonical rasters; apparent motion shows frame 1, with sequence facts retained in source data.

Reading guide: luminance and raster levels describe pixels; frame rate and frame delta describe temporal media; neither is an observer report.

visual.duck_rabbit



canonical raster

Luminance mean: 0.777 - SD: 0.371

Levels: 3 - digest: dde9d5ecda...

visual.simultaneous_contrast



canonical raster

Luminance mean: 0.500 - SD: 0.328

Levels: 3 - digest: 696ba38b0a...

visual.apparent_motion

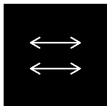


representative first frame; sequence in source data

Frames: 6 - rate: 12.0 frames/s

Frame delta: 0.242 normalized - digest: acc31937b4...

visual.muller_lyer



canonical raster

Luminance mean: 0.027 - SD: 0.162

Levels: 2 - digest: 3e97a33a27...

visual.poggendorff



canonical raster

Luminance mean: 0.989 - SD: 0.106

Levels: 2 - digest: 21d6b959d4...

visual.ponzo



canonical raster

Luminance mean: 0.958 - SD: 0.201

Levels: 2 - digest: e5a04c8921...

visual.kanizsa_triangle



canonical raster

Luminance mean: 0.921 - SD: 0.270

Levels: 2 - digest: db5f2b297b...

visual.ebbinghaus



canonical raster

Luminance mean: 0.347 - SD: 0.152

Levels: 3 - digest: 1560d41307...

visual.zollner



canonical raster

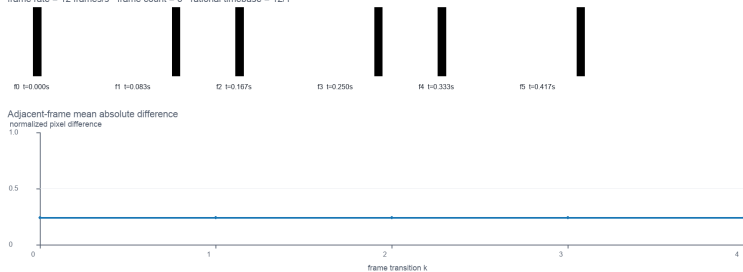
Luminance mean: 0.937 - SD: 0.242

Levels: 2 - digest: 917daad5dc...

Temporal and auditory constructions I

Temporal sequence and frame deltas

frame rate = 12 frames/s · frame count = 6 · rational timebase = 12/1



Frame timestamps and deltas are decoded/canonical timing facts; apparent motion remains an observer-level hypothesis.

Figure 5: Six ordered frames are labeled with timestamps and paired with a line plot of adjacent-frame normalized pixel differences.

Temporal and auditory constructions II

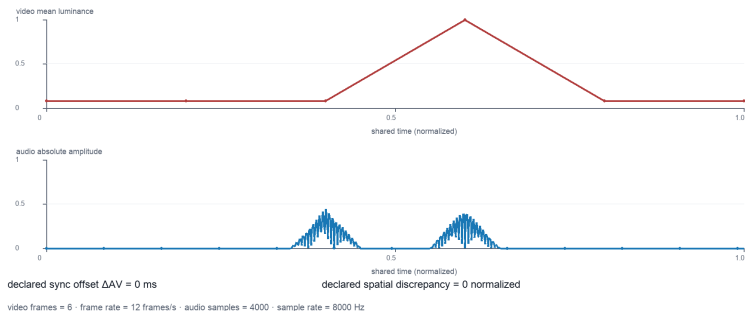
What this figure shows: Six ordered frames are labeled with timestamps and paired with a line plot of adjacent-frame normalized pixel differences. The frame strip exposes the order and timestamps of the apparent-motion construction, while the lower trace reports adjacent-frame mean absolute pixel differences.

Source data `output/data/temporal_sequence.json` preserve frame count, frame rate, reduced rational timebase, timestamps in seconds, and the nonzero transition series; seed 0 identifies the generated sequence. Controls: `visual.apparent_motion` default typed parameters; frame rate and frame count from the canonical sequence; seed `s=0`. Objective facts: timestamps in seconds, frame rate in frames/s, frame count, rational timebase, and adjacent-frame normalized pixel difference. Claim level: `physical_metric`. Source data: `output/data/temporal_sequence.json` (SHA-256 digest recorded in the figure registry). Evidence lineage: `wertheimer1912motion`, `sekuler1996wertheimer`. Limitations: Frame succession and frame deltas are physical file properties; playback timing, display persistence, and motion reports require a controlled observer task.

Audiovisual timing and encoding I

Audiovisual shared-clock timeline

Video and audio are plotted against normalized shared time; offsets are declared, not inferred from perception.



The aligned traces verify physical timing metadata; audiovisual capture or temporal recalibration requires a controlled observer task.

Figure 7: Two labeled traces share a normalized time axis: video mean luminance and audio absolute amplitude, with signed synchronization and spatial offsets printed below.

Audiovisual timing and encoding II

What this figure shows: Two labeled traces share a normalized time axis: video mean luminance and audio absolute amplitude, with signed synchronization and spatial offsets printed below. Video mean luminance and rectified audio amplitude are placed on a common normalized time axis for the sound-induced-flash construction. Source data output/data/audiovisual_timeline.json records the video frame clock, audio sampling clock, event traces, signed audio-minus-video synchronization offset in milliseconds, and normalized spatial discrepancy; alignment is declared by the generator, not estimated from perception. Controls: audiovisual.sound_induced_flash default shared timeline; declared sync offset ΔAV and spatial offset; seed $s=0$. Objective facts: video luminance and audio envelope on normalized shared time; ΔAV in ms; spatial discrepancy in normalized units; frame and sample clocks in the sidecar. Claim level: physical_metric. Source data: output/data/audiovisual_timeline.json (SHA-256 digest recorded in the figure registry). Evidence lineage: shams2000sifi, hirst2020sound, vroomen2004temporal, hartcherobrien2011temporal. Limitations: Display refresh, audio

Objective metric results I

Table 3: Objective metric definitions and epistemic boundary.

Metric	Unit	Definition	Tolerance	Claim level
mean_luminance	normalized_luminance	mean Rec. 709 relative luminance (grayscale is identity)	exact canonical	physical_metric
unique_values	levels	count of distinct canonical image values	exact integer	physical_metric
rms	normalized_amplitude	root-mean-square audio amplitude	finite scalar	physical_metric
peak	normalized_amplitude	maximum absolute audio amplitude	finite scalar	physical_metric
spectral_centroid_hz	Hz	energy-weighted one-sided spectral centroid	finite scalar	physical_metric
spectral_bandwidth_hz	Hz	spectral standard deviation around the one-sided centroid	finite scalar	physical_metric
crest_factor	ratio	audio peak divided by RMS with zero-RMS guard	finite scalar	physical_metric
mean_temporal_delta	normalized_pixel_difference	mean adjacent-frame absolute difference	finite scalar	physical_metric
sync_offset	ms	declared audio-minus-video offset	typed value	physical_metric

Objective metric results II

The metric definitions in `[@tbl:metrics]` are computed from canonical or decoded artifacts and do not encode observer interpretations.

The default duck-rabbit artifact contains 3 unique raster levels and has mean normalized luminance 0.7770. These values are live generated facts, not perceptual effect sizes.

Observer estimands and verification controls I

Table 4: Data-free observer outcomes and preregistered estimands.

Estimand	Response and model	Reference and contrast	Uncertainty
sifi flash count contrast	event count; poisson or ordinal	sifi one beep; two-beep minus one-beep event-count response	95% confidence interval
temporal binding offset contrast	continuous magnitude; linear mixed	temporal sync; reported timing shift per declared sync offset	95% confidence interval

Observer estimands and verification controls II

The analysis contract in `[@tbl:observer_estimands]` specifies estimands and uncertainty procedures without asserting any result.

Table 5: Verification failure modes and negative controls.

Failure mode	Control	Expected result
altered encoded file	recompute encoded SHA-256	fail
stale canonical digest	recompute canonical little-endian float32 digest	fail
incorrect decoded dimensions	compare inspection facts with manifest summary	fail
stream timing mismatch	compare duration/timebase and declared sync offset	fail
unsupported backend	capability probe before encoding	explicit capability error
conflicting format requests	reject format_name/output_spec disagreement	parameter error

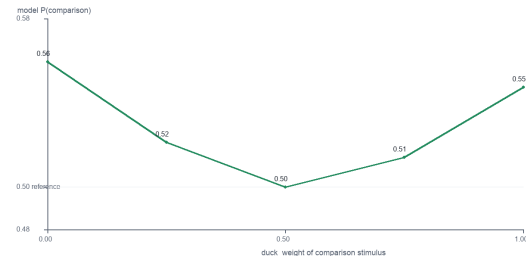
Observer estimands and verification controls III

The negative controls in [tbl:verification] test package integrity and media contracts; they are not null findings about perception.

Synthetic psychophysics model diagnostic I

Synthetic psychophysics: transparent feature observer

A deterministic model diagnostic varies duck_weight against a fixed reference; it is not a human or pretrained vision-model result.



Model-output scale: 0.48–0.58 · reference = 0.50 · human_data: false · training_data: none

Model contract

```
model = duckrabbit.synthetic.feature_observer
version = 1.0
temperature = 0.05
training data = none
calibration = analytic logistic
human validation = not performed

reference = duck_weight=0.50
points = 5
P range = 0.500–0.559
```

This curve diagnoses sensitivity of a specified feature observer. It does not estimate a psychometric function or establish a perceptual effect.

Figure 9: A narrow-range line plot shows deterministic model probability across duck_weight values beside the model identity, temperature, no-training-data statement, and human-data boundary.

Synthetic psychophysics model diagnostic II

What this figure shows: A narrow-range line plot shows deterministic model probability across duck_weight values beside the model identity, temperature, no-training-data statement, and human-data boundary. The hand-specified feature observer compares duck_weight variants with a fixed canonical reference. Its probability trace is computed analytically from serialized features, weights, and temperature; source data output/data/synthetic_psychophysics.json records model ID/version, reference and comparison digests, seed, calibration, training_data=none, and human_data=false. The y-axis is deliberately a narrow model-output range rather than a human psychometric scale. Controls: duck_weight comparison against fixed reference; serialized feature weights and temperature; seed s=0. Objective facts: model probability and score delta on the displayed model-output scale; canonical stimulus digests; human_data=false; training_data=none. Claim level: synthetic_model_output. Source data: output/data/synthetic_psychophysics.json (SHA-256 digest recorded in the figure registry). Evidence lineage: